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8 February 1950
SUPERSEDING
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16 December 1943

MILITARY SPECIFICATION

TAPE, SILK

(FOR USE IN LOADING OF AMMUNITION)

1. SCOPE AND CLASSIFICATION.

1.1 This specification covers one grade and the two following classes of silk tape for use in loading of ammunition:

Class 1 - 1/4 inch width.
Class 2 - 1/2 inch width.

2. APPLICABLE SPECIFICATIONS.

2.1 Specifications.- The following U. S. Army specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

50-0-1 - General Specification For Ammunition Except Small Arms Ammunition.
100-2 - Standard Specification For Marking Shipments By Contractors.

(Army.--Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS.

3.1 Material.-- The tape shall be manufactured from pure silk without the admixture of any other material whatever, except the amount of sizing absolutely necessary to facilitate weaving.

3.2 Width.--

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3.2.1 Class 1.-- 0.28 inch maximum, .20 inch minimum.
3.2.2 Class 2.-- 0.50 inch maximum, 0.35 inch minimum.
3.3 Weight per yard.--
3.3.1 Class 1.-- .035 ounce, maximum.
3.3.2 Class 2.-- 0.05 ounce, maximum.
3.4 Breaking strength.--
3.4.1 Class 1.-- Minimum, average, 50 pounds, with no specimen breaking at less than 35 pounds.
3.4.2 Class 2.-- Minimum, average, 40 pounds, with no specimen breaking at less than 35 pounds.
3.5 Elongation.-- Maximum, 34 percent.
3.6 Objectionable sizing.-- The character and quality of the sizing materials shall be such that there is no indication of smoldering when the tape is ignited sufficiently to give a good flame and then extinguished. The tape shall pass the objectionable sizing test.
3.7 Ether extract.-- The ether extract shall be not more than 10 percent.
3.8 Acidity or alkalinity.--
3.8.1 pH of water extract.- The pH of the water extract of the tape shall be not less than 4.4 nor more than 9.7.
3.8.2 Organic acidity or alkalinity.-- The water extract of the tape shall show not more than 0.1 percent acidity, calculated as acetic acid, nor more than 0.1 percent alkalinity, calculated as sodium carbonate.
3.9 Ash.-- Maximum, 2.0 percent.
3.10 Workmanship.-- The tape shall be of a twill weave and shall have a selvage along each edge. It shall be even in texture and free from lumps, knots, or other irregularities.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.
4.1 Sampling.--
4.1.1 Size of lot.-- A lot shall consist of 50,000 yards, maximum.
4.1.2 Sampling procedure.- Ten percent but in no case more than 10 nor less than 5 of the packages comprising the lot shall be selected by the inspector so as to be representative of the lots except that if the lot consists of less than 5 packages, each package shall be sampled. Remove from each selected package 1 roll of the material. Discard 1 complete turn from each selected roll, and cut a 2-yard sample from the roll. Label these pieces to show the name of the material, manufacturer, plant, purchase order, and number of yards in the lot. All acceptance tests shall be made on the composite sample representative of the lot.

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4.2 Inspection.—Inspection shall be made in accordance with U. S. Army Specification No. 50-O-1 and shall be made at the point of delivery, unless otherwise specified in the contract or order.

4.3 Tests.— The laboratory tests shall be made in accordance with the following paragraphs:

4.3.1 Width.—Measure at two places the width of each of the pieces in the composite sample, using a steel rule accurate to 0.01 inch. Calculate the average of the measurements.

4.3.2 Weight.— Weigh at least 5 of the pieces in the composite sample to the nearest 0.01 gram and measure the length of each weighed piece by means of a steel rule or measuring stick accurate to 0.1 inch. Calculate the average weight in ounces per yard.

4.3.3 Breaking strength.—Cut one or more test specimens, 24 inches in length, from each piece of material comprising the composite sample. Test each of these specimens by means of a pendulum type machine of such capacity that the arm makes an angle between 9° and 45° with the vertical when the specimen breaks. The machine shall have a rate of separation of the grips of 12 ± 2 inches per minute under no load. Use smooth grips of the drum type, 2 inches in diameter. At the beginning of the test the distance between the grips shall be not less than 6 inches. Calculate the average breaking strength.

4.3.4 Objectionable sizing.—

4.3.4.1 Apparatus.— Prepare a Mackey test apparatus (see Figure 1) consisting of a cylindrical water jacketed metal oven of the following dimensions: Outside, 8 inches high and 6 inches in diameter; inside, 7 inches high and 4 inches in diameter. Seal the vessel with a lid lined with nonconducting material having three holes, one at the center for a thermometer, and two diametrically opposite near the rim. These holes receive copper tubes of ½-inch diameter so arranged that when the lid is in place, one tube enters the oven to a depth of 6 inches while the other rises to an equal height above the lid. These tubes assure a constant draft of air through the instrument. On a common vertical axis with the central hole, a cylinder of wire gauze, 6 inches long and 1½ inches in diameter, shall be supported within the oven.

4.3.4.2 Procedure.—Place 12 ± 1 gm of the sample cut into ¼-inch to ½-inch squares within the wire cylinder, compressing it to occupy the upper 4½ inches. Arrange the thermometer so that the bulb is in the center of this mass. Heat the water to the boiling point, then introduce the cylinder and thermometer, allowing the latter to protrude through a cork placed in the central hole in the lid. Allow the water to boil and record the temperature at frequent intervals for 1½ hours. Consider the material to fail with respect to the presence of objectionable sizing if the fabric attains a temperature of 100°C. within 1 hour, or it reaches a temperature of 120°C., within 1½ hours.

4.3.5 Ether extract.—Cut 10 pieces of tape, weighing approximately 0.5 gm each, from various portions of the sample. Weigh these pieces together and transfer them to the extraction chamber of a Soxhlet extraction apparatus having ground-glass connections. Assemble the extraction chamber to a tared receiving flask and transfer ether to the extraction chamber until the sample is covered and the ether siphons over into the flask.

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Add approximately 25 ml more ether to the extraction chamber, assemble a condenser to it and place the assembled apparatus on a steam bath. Allow the extraction to proceed for 1.5 hours. Discontinue heating at a time when the ether content of the extraction chamber is nearly sufficient to cause the ether to siphon over into the receiving flask. Heat the flask, containing the oil ether extract, on a steam bath until the ether is removed. Dry the flask and residue in an oven at 100°C. ± 2°C., for 1 hour, cool in a desiccator and weigh. Calculate the weight of residue to percent ether extract in the sample.

4.3.6 Acidity or alkalinity.—

4.3.6.1 pH of water extract.—Cut 10 pieces of tape, weighing approximately 0.5 gm each, from various portions of the sample. Combine these pieces, weigh, and transfer to a 250 ml beaker. Add 150 ml of freshly boiled, distilled water to the beaker, cover with a watch glass, and boil the water gently for 30 minutes. In case the volume of water becomes less than one-half the original amount during the boiling period, add sufficient boiling distilled water to restore it to the original volume. At the end of 30 minutes remove the covered beaker and contents from the source of heat and bring the volume to 150 ml by adding freshly boiled distilled water. Cool the covered beaker and contents to 25° to 30°C. without agitating the contents, to minimize the absorption of carbon dioxide from the atmosphere. Determine the pH of the cooled liquid using a glass electrode pH electrometer within 1 hour after removing the beaker from the source of heat. The distilled water used for this test should be carried through the above operations as a blank, and must show a pH value in the range 6.0 to 7.0.

4.3.6.2 Organic acidity or alkalinity.—Filter the solution used in paragraph 4.3.6.1. If the pH of the solution is between 7.0 and 9.7, titrate the solution with N/10 hydrochloric acid solution using methyl red as the indicator. If the pH of the solution is between 7.0 and 4.4, titrate with N/10 sodium hydroxide using phenolphthalein as the indicator. Calculate any acidity to percent acetic acid and any alkalinity to percent sodium carbonate.

$$\text{Acidity, as acetic acid percent} = \frac{A \times B \times 6.005}{W}$$

where: A = ml of sodium hydroxide used in titration
B = normality of sodium hydroxide used in titration
W = weight of original sample

$$\text{Alkalinity, as sodium carbonate, percent} = \frac{AxB \times 5.300}{W}$$

where:
A = ml of hydrochloric acid used in titration
B = normality of hydrochloric acid used in titration
W = weight of original sample

4.3.7 Ash.—Cut parts from each of the pieces comprising the sample so as to obtain a total of approximately 2 gm and transfer to a tared platinum or porcelain crucible and moisten with concentrated nitric acid. Heat

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the crucible on a steam bath for 1 hour and then heat carefully over a flame so as to avoid any loss by spattering. When the contents of the crucible have been charred, heat to dull redness until all carbonaceous matter has been burned off. Cool the crucible in a desiccator and weigh. Calculate the gain in weight to percent ash in the sample.

4.3.8 Elongation.—The elongation of the tape shall be obtained when the breaking strength is determined by means of a suitable autographic recording device on the testing machine. An initial load of 6 ounces shall be placed on the specimen before gripping it in the lower jaws of the machine. The elongation shall be the average of the results obtained for 5 specimens. It shall be calculated from the start of the line as shown on the graphic record and expressed as the percentage increase in length.

4.4 Rejection.—If a sample fails to meet any of the requirements or tests listed in this specification, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY.

5.1 Packaging.—Unless otherwise specified, commercial packages, containing not more than 200 yards of tape, are acceptable under this specification.

5.2 Packing.—Unless otherwise specified, silk tape shall be delivered in standard commercial containers so constructed to insure acceptance by common or other carriers, for safe transportation, at the lowest rate to the point of delivery.

5.3 Marking.—Marking for shipment shall be in accordance with U. S. Army Specification No. 100-2.

6. NOTES.

6.1 NOTICE:—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

Custodian:

Army—Ordnance Department

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MACKEY TEST APPARATUS

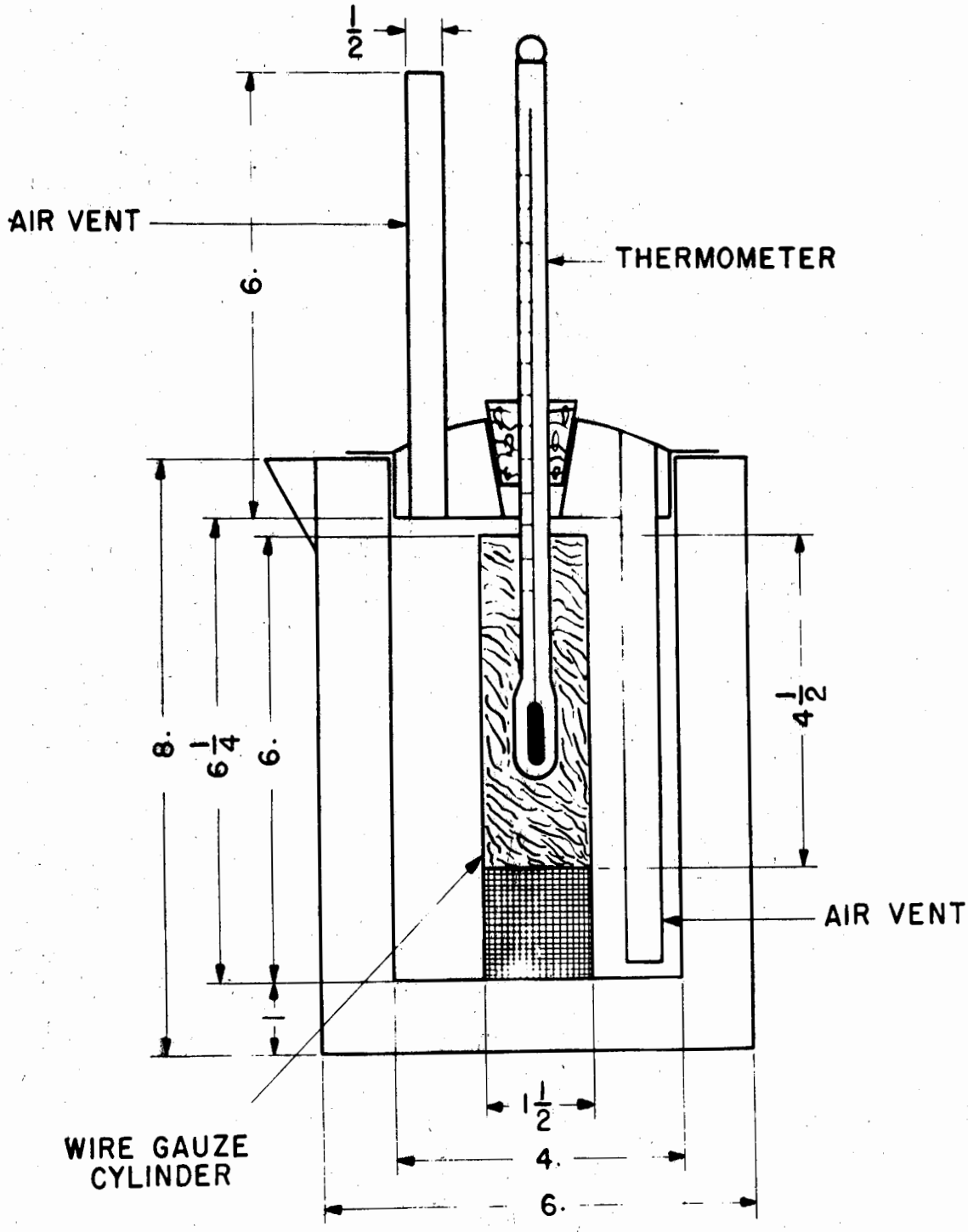


FIG. 1

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